

Report Guidelines

Applied Machine Learning & Data Science Projects

Data Science 435 / 535 & Computer Science 449 / 549 • Fall 2026

Initial & Midterm Reports:

There will be an initial and midterm report due during the semester to provide guidance and feedback on each team's progress. The initial report should include an introduction to the project with project background and objectives, a description of the data, and the data science pipeline design plan. The midterm report should additionally include any progress on data wrangling, data exploration, and modeling.

Final Report:

The final report is due to the instructors and the project sponsor at the end of the semester and is a major component of the course grade.

Possible Report Sections:

Loosely, the report should consist of the following sections. Note that the precise organization / sections of the report are up to the team, but each report should at least contain this content:

1. Introduction:

- The introduction should be written for a general audience and should outline the problem and its significance.
- Background: Include background research on the problem / problem area with a literature review. This includes background on the technical approaches and techniques that others have used to address similar problems.
- Objectives: State the project objectives clearly.
- Data Science Pipeline: Give an overview or outline of your data science pipeline design to address the objectives.

2. Data & Data Exploration:

- Data Description: Fully describe all data used for the project.
- Data Wrangling: Describe any wrangling done to format the data for analysis.
- Feature Engineering / Feature Filtering: If any features were engineered or featured filtered, describe the methods used, the resulting features, and why.
- Exploratory Data Analysis & Visualization: Include a full exploratory data analysis and visual summaries of the data.

3. Methods & Models:

- Fully describe all data science techniques and models used in the project.

4. Results:

- Include all project results and explain how to interpret the results in a manner appropriate for a general audience.
- Include a validation plan and validation of all key results.

5. Conclusion:

- Conclude by discussing the implications and potential future work related to your project.

Report Grading Rubric:

1. Communication: 30%

- Is the written communication professional (e.g. no spelling or grammar mistakes)? (Note that ANY spelling or grammar mistake will automatically result in a 10% reduction in the grade.)
- Visualizations. Are the graphics clear, engaging, informative, and appropriate to convey the results? Are all your texts in figures, e.g., axes labels, legends, and other texts, of adequate font size to be legible?
- Is the report well-organized and clear?
- Is the introduction, interpretation of results, and conclusions communicated clearly for a general audience?

2. Significance: 10%

- What is the significance and impact of this work in both the applied discipline and in data science?

3. Innovation & Creativity: 10%

- Did you employ creative or innovative solutions? Apply data science techniques in a novel way? Or develop or extend new data science techniques?

4. Technical Approach: 35%

- Are sound data science techniques used at each step along the data analysis pipeline? Are techniques and pipeline choices reasonable and justified?

5. Data Scientific Reliability & Reproducibility: 15%

- Are best practices and techniques used to validate all data analysis findings?
- Are all results likely to be reproducible and reliable?

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Report Grading Checklist:

- Communication (30% of grade):
 - Adherence to professional writing standard:
 - no spelling mistakes
 - no grammar mistakes
 - no colloquial language
 - no acronyms without spelling out for the first time
 - Organization:
 - Report is complete with all required content
 - Report is well-organized into proper sections and subsections
 - Report has a table of contents and page numbers
 - Communication is appropriate for targeted audience:
 - introduction, interpretation of results and conclusions should be targeting for lay audience (e.g. high school students)
 - technical approaches should be targeting for audience with general data science knowledge (e.g. your classmates)
 - Raw variable names have been avoided as much as possible
 - Visualizations:
 - Good balance of texts and visualizations
 - Graphs and plots are appropriately used
 - Graphs and plots are professional with high resolution
 - Plots have titles and captions
 - All labels and ticks are legible (not too small)
 - All labels and ticks are meaningful (no raw names or codes)
 - Visualizations are informative and engaging
 - Captions:
 - All figures and tables have captions
 - All captions are self-explanatory
 - Literature Review:
 - Literature relevant to the problem and domain area is appropriately cited and discussed
 - Literature on related technical approaches are cited and discussed
 - All references are properly cited

Comments:

- Technical Approach (35% of grade):
 - Data science pipeline design plan is clearly laid out
 - Sound data science techniques are used at each step along the data science pipeline
 - Data science technical choices are reasonable and justified
 - Data science technical choices are fully described in enough detail to reproduce the analysis
 - Modeling assumptions are stated and checked appropriately
 - Models are properly trained and tuned
 - Results are reported clearly and interpreted in context
 - Performance measures are reported and compared to baseline model if appropriate

Comments:

- Reproducibility & Reliability (15% of grade):
 - Data is properly splitted into training, validation and testing, if applicable
 - All the analysis and modelling results are validated
 - Results are reproducible with the same data
 - Data science pipeline is reusable with new data
 - Results and conclusions are reliable and lead to actionable findings

Comments:

- Impact & Innovation (20% of grade):
 - Discussion of impact of this work in the applied discipline
 - Discussion of impact of this work in data science
 - Employment of creative and innovative solutions to the real data problems
 - Extension of existing data science or machine learning techniques
 - Development of new data science or machine learning methodologies

Comments:

Overall Grade:

Overall Comments: